

# Hypergraph Representations for Noisy Plant Taxonomy, Reticulate Evolution, and Hierarchy-Aware Biological Classification

2026-05-17

## Contents

|                                                                                                                                 |          |
|---------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Hypergraph Representations for Noisy Plant Taxonomy, Reticulate Evolution, and Hierarchy-Aware Biological Classification</b> | <b>1</b> |
| Abstract . . . . .                                                                                                              | 1        |
| Introduction And Research Question . . . . .                                                                                    | 2        |
| Evidence Base And Data Feasibility . . . . .                                                                                    | 3        |
| Hypergraph Formulation And Baselines . . . . .                                                                                  | 4        |
| Synthetic Benchmark And Hierarchy-Aware Metrics . . . . .                                                                       | 5        |
| Public WFO/GBIF/Open Tree Source-Plumbing Sample . . . . .                                                                      | 6        |
| Baseline, Hypergraph, And Ablation Results . . . . .                                                                            | 7        |
| Formal Clique-Expansion Diagnostic . . . . .                                                                                    | 8        |
| Claim Classification And Hypothesis Assessment . . . . .                                                                        | 10       |
| Limitations, Residual Debt, And Future Work . . . . .                                                                           | 11       |
| Reproducibility And Artifact Map . . . . .                                                                                      | 12       |
| Conclusions . . . . .                                                                                                           | 15       |
| References . . . . .                                                                                                            | 15       |

## Hypergraph Representations for Noisy Plant Taxonomy, Reticulate Evolution, and Hierarchy-Aware Biological Classification

### Abstract

This campaign tested whether plant taxonomic hierarchy, synonymy, trait similarity, geography, and reticulate evolutionary signals can be represented as a hypergraph in a way that gives a measurable advantage over tree, directed acyclic graph (DAG), ordinary graph, or flat-classification baselines. A hypergraph is a graph-like structure in which one edge can connect more than two nodes. That property matches several plant-taxonomy data situations: a synonym cluster can link many names to one accepted taxon, a source-context record can link a name, rank, identifier, source, and region, and a reticulate or hybrid-like signal can involve more than one source lineage.

The result is a scoped modeling and diagnostic contribution, not a claim of broad predictive superiority. The campaign produced a role-labeled hypergraph formulation, deterministic synthetic benchmark, hierarchy-aware metric stack, small public WFO/GBIF/Open Tree source-plumbing sample, deterministic baseline and ablation suite, and formal clique-expansion diagnostic. The final audit recorded all eight planned milestones as validated with high confidence, no critical, moderate, or minor findings, no residual debt, and `promise_check=green`.

The central empirical finding is mixed and useful. In the validated synthetic experiment, native hypergraph scoring improved mean hierarchy distance over tree/DAG, ordinary graph,

and clique expansion, but it did not improve flat exact accuracy. The main contribution is therefore not a state-of-the-art classifier. It is a reproducible artifact that makes the hypergraph question testable while preserving the distinctions among taxonomy, phylogeny, nomenclature, trait similarity, occurrence evidence, and dataset labels.

The report keeps all biological claims bounded. Public data supported name, identifier, status, taxonomy, and source-context plumbing only. Reticulate or hybrid-like structure, trait convergence, missing-rank stressors, and noisy occurrence labels were synthetic benchmark mechanisms, not claims about real plant hybrid origins, species relationships, traits, ranges, or phylogenetic novelty.

## Introduction And Research Question

The research question was:

Can plant taxonomic hierarchy, synonymy, trait similarity, geography, and reticulate evolutionary signals be represented as a hypergraph in a way that produces a measurable advantage over tree, DAG, ordinary graph, or flat-classification baselines?

The campaign was designed as an artifact-building effort rather than a literature review. Its target reader is a computational biology, biodiversity informatics, or machine-learning researcher deciding whether hypergraph modeling is worth pursuing for plant classification and taxonomic conflict analysis.

Plant taxonomy is hierarchical in form, but plant data systems are not clean trees in practice. They must handle accepted names, synonyms, unresolved names, missing intermediate ranks, disputed circumscriptions, source-specific checklist behavior, geographic and occurrence context, noisy labels, hybrids, cultivars, and evidence layers that do not all mean the same thing. A strict hierarchy can represent parent-child rank structure, but it cannot by itself express a multi-way synonym cluster, an occurrence provenance record, a source-specific checklist context, or a reticulate signal involving multiple source lineages.

The report uses these terms consistently:

- **Taxonomy** is a managed classification with ranks, accepted names, synonyms, and circumscriptions.
- **Phylogeny** is inferred evolutionary relationship or synthesis evidence. It may be tree-like, uncertain, or network-like, but it is not identical to a managed taxonomy.
- **Nomenclature** is the system of scientific names, authorship, synonymy, and name-status records.
- **Trait similarity** is morphological, ecological, physiological, or phenological resemblance. It can support candidate similarity but does not prove accepted taxonomy or phylogeny by itself.
- **Occurrence evidence** is specimen or observation evidence with geography, date, source, and quality fields.
- **Dataset label hierarchy** is the label structure available in a particular training or benchmark dataset. It may be incomplete or inconsistent with a full taxonomic authority.

The modeling baselines follow the same separation. A **tree/DAG baseline** uses taxonomic parent-child hierarchy only. A **DAG**, or directed acyclic graph, is a directed graph with no directed cycles; in this campaign it represents accepted hierarchy structure without native multi-way evidence. An **ordinary graph baseline** represents evidence through pairwise edges such as parent-child, synonym-of, trait-similar-to, or source-context links. A **flat baseline** predicts labels without using hierarchy. A **native hypergraph model** keeps role-labeled

multi-way incidence, meaning one hyperedge can connect several typed nodes while preserving each node's role in that relation.

This distinction matters because the safe research target is modeling, diagnostics, and benchmark construction. The campaign avoided claims that would require authoritative botanical or phylogenetic evidence beyond the collected artifacts. It did not claim new plant taxonomy, corrected accepted names, species relationships, hybrid origins, trait syndromes, species ranges, checklist truth, occurrence-quality conclusions, or phylogenetic novelty.

## Evidence Base And Data Feasibility

The campaign first established which evidence layers could be used reproducibly. The data feasibility map in `docs/data_feasibility_map.md`, its structured counterpart `data/source_feasibility_matrix.csv`, and the probe output `data/source_probe_results.json` supported a narrow no-auth public-data path. That path used World Flora Online (WFO), GBIF, and Open Tree of Life for small name, taxonomy, identifier, and source-context samples. It did not use public data for real reticulation, trait, occurrence-quality, range, or phylogenetic-novelty claims.

WFO was selected as the primary plant taxonomy and nomenclature source for the first prototype. The relevant WFO materials included the Plant List API, name-matching REST API, GraphQL documentation, downloadable Plant List snapshot, and Plant List description [1]-[5]. In the campaign artifacts, WFO supported accepted plant names, candidate name matching, synonyms, ranks, WFO identifiers, and source-specific taxonomic backbone context.

GBIF was treated as a separate biodiversity evidence system rather than a substitute for WFO. Its API documentation, species API, occurrence API, download guidance, citation guidelines, and terms of use were recorded as source references [6]-[11]. The feasible first-pass role for GBIF was taxon matching and small interactive source checks. The campaign did not make authenticated GBIF occurrence downloads part of the reproducible path, and it did not use GBIF occurrence records to validate geography or occurrence-quality claims.

Open Tree of Life was treated as taxonomy/phylogeny synthesis context, not as an authority identical to WFO or GBIF. The campaign recorded the Open Tree API, TNRS API, licensing, and synthesis-tree reference [12]-[15]. In the public sample, Open Tree contributed name-resolution and OTT identifier context. It did not turn the sample into a public phylogenetic analysis.

BIEN/RBIEN and TRY were assessed but not used as required inputs for the first reproducible artifact. BIEN/RBIEN remained optional because normal access is mainly a local R/package setup issue and was not needed for the narrow prototype [16]-[17]. TRY was treated as access-limited for this campaign because registration, request, citation, intellectual-property, and redistribution constraints made it unsuitable for a no-auth redistributable first benchmark [18]-[20], even though TRY is an important plant trait resource [26].

The feasibility work also recorded method literature for the modeling side. Hypergraph learning references supported the idea that multi-way relationships can be used for clustering, classification, and embedding [21]-[22]. Hierarchical classification literature supported evaluating predictions by hierarchy-aware criteria rather than flat correctness alone [23]. Biodiversity knowledge-graph literature supported preserving source-specific biological data relationships rather than flattening everything into one reconciled table [24]-[25].

The practical decision from this evidence map was a hybrid data strategy. Public data would support real name, identifier, status, rank, taxonomy, and source-context plumbing. Synthetic data would support controlled reticulation, missing-rank, trait-convergence,

noisy-occurrence, and benchmark-label mechanisms. This split was the campaign’s main protection against overstating biology: public data made the schema realistic at the name/taxonomy/source layer, while synthetic mechanisms supplied known ground truth for falsification-oriented model tests.

The final audit confirmed that all eight planned milestones were validated with high confidence. For this evidence-base section, the validated milestone anchors were M1 for data feasibility, M2 for the schema, M3 for the synthetic benchmark, M4 for metrics, and M5 for the public source sample. Later sections use M6 and M7 for model results and the formal diagnostic.

## Hypergraph Formulation And Baselines

The formal representation was a typed, weighted hypergraph:

$$H = (V, E, X, W).$$

Here,  $V$  is the set of typed nodes,  $E$  is the set of typed hyperedges,  $X$  stores optional node or edge features, and  $W$  stores edge weights and provenance confidence. An incidence matrix records participation:  $B[v, e] = \text{role\_weight}(v, e)$  when node  $v$  participates in hyperedge  $e$ . The role weight matters because a source node, accepted taxon node, synonym name node, and region node may all occur in the same hyperedge without meaning the same thing.

The schema defined node types for taxon, name\_string, rank, trait, region, occurrence\_record, source, specimen\_or\_observation, phylogeny\_node, and dataset\_label. The point of this typing was to prevent one layer from silently standing in for another. A WFO taxon identifier, a GBIF taxon key, an Open Tree OTT identifier, a synthetic trait state, and a benchmark label can all participate in the same representation, but their node types preserve the distinction among taxonomy, nomenclature, phylogeny, trait evidence, source provenance, and dataset labels.

Seven hyperedge families formed the binding schema:

| Hyperedge family           | Allowed use                                                                                    | Forbidden inference                                                                     |
|----------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| taxonomic_parentage        | Hierarchy consistency, ancestor/descendant paths, and hierarchy-distance evaluation.           | It does not prove phylogeny or trait similarity by itself.                              |
| synonym_cluster            | Name normalization and synonym-aware penalties.                                                | It does not prove biological closeness beyond accepted-name resolution.                 |
| trait_syndrome             | Trait-based candidate evidence and convergence stress tests.                                   | It does not prove accepted taxonomy, phylogeny, or real trait syndromes in public data. |
| regional_checklist_context | Source- and region-scoped taxonomic context.                                                   | It does not prove that checklist differences are biological conflicts.                  |
| occurrence_provenance      | Label provenance, geographic plausibility, source-specific noise, and occurrence stress tests. | It does not prove ranges or occurrence quality without authoritative evidence.          |

| Hyperedge family                         | Allowed use                                                | Forbidden inference                                                                    |
|------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <code>reticulate_or_hybrid_signal</code> | Multi-parent near-miss scoring and tree-obstruction tests. | It does not prove real hybrid origins unless backed by independent biological sources. |
| <code>missing_rank_bridge</code>         | Label completion when an intermediate rank is absent.      | It does not assert that the missing rank is biologically nonexistent.                  |

The baselines were defined by information budget. A flat baseline could use label frequencies but no hierarchy. A taxonomy-depth baseline could exploit rank depth or common hierarchy depth without using full source evidence. A tree/DAG baseline could use accepted parent-child hierarchy. An ordinary graph baseline could use pairwise edges for relationships such as parent-child, synonym-of, trait-similar-to, or source-context links. A clique-expansion baseline converted each multi-way hyperedge into pairwise graph edges. The native hypergraph model kept the original multi-way incidence and role labels.

This baseline design made the comparison falsifiable. The hypergraph was not allowed to win merely because it saw evidence hidden from the baselines. The ordinary graph and clique-expansion baselines were given access to structurally comparable evidence after conversion to pairwise edges, while the tree/DAG baseline represented the narrower taxonomic hierarchy-only condition. Later experiments could therefore distinguish three possibilities: no structural gain, a gain caused mostly by synonym or label-frequency effects, or a hierarchy-aware gain linked to native multi-way incidence.

## Synthetic Benchmark And Hierarchy-Aware Metrics

The deterministic synthetic benchmark was built to supply known ground truth for mechanisms that public data could not safely provide in a first reproducible artifact. The generator `scripts/generate_synthetic_benchmark.py` writes `data/synthetic_benchmark/v0.1/` with `taxa.csv`, `names.csv`, `hyperedges.csv`, `examples.csv`, `splits.csv`, `metadata.json`, and `composition.png`. The design note `docs/synthetic_benchmark_design.md` records the generation logic.

The generated benchmark contained 76 taxa, including 60 species; 138 names; 1081 hyper-edge incidence rows; and 103 examples. The split was 62 training examples, 19 validation examples, and 22 test examples. The benchmark included 7 reticulate cases and 12 trait-convergence taxa. Its examples covered strict hierarchy, synonym or renamed labels, missing intermediate ranks, trait convergence, noisy occurrence labels, and reticulate or hybrid-like cases.

The benchmark’s reticulation claim was deliberately structural. In the synthetic data, a strict single-parent tree projection must either drop one source lineage or duplicate a child when the intended label structure contains multiple source lineages. The native incidence representation can keep both source-lineage roles in one `reticulate_or_hybrid_signal` hyperedge. This does not assert a real plant hybrid origin; it creates a controlled finite case where the representational cost of a single-parent tree can be tested.

The benchmark also included a strict negative-control mode. In that mode, reticulation, trait convergence, missing ranks, occurrence noise, and synonym or rename mechanisms are disabled. This creates a tree-generated setting where a tree/DAG baseline should not be structurally disadvantaged. That control is central to the campaign’s falsification design: if the hypergraph still appeared to gain in a purely tree-like dataset, the result would require scrutiny

for leakage, label-frequency effects, or scoring artifacts.

The metric stack was implemented in `tools/hierarchy_metrics.py` and validated by `tests/test_hierarchy_metrics.py` and `tests/test_synthetic_benchmark.py`. It included:

| Metric                                          | What it measures                                                                                                    |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <code>flat_exact_match</code>                   | Whether the predicted label exactly matches the target label.                                                       |
| <code>synonym_normalized_exact_match</code>     | Whether prediction and target collapse to the same accepted taxon under the benchmark's synonym mapping.            |
| <code>hierarchy_distance</code>                 | The path-based hierarchy error between prediction and target in the accepted hierarchy, with missing-rank handling. |
| <code>mean_hierarchical_distance_error</code>   | Mean hierarchy distance over a set of examples.                                                                     |
| <code>hierarchy_coherence_violation_rate</code> | Frequency of multi-rank predictions whose family, genus, and species labels cannot coexist after normalization.     |
| <code>reticulate_near_miss_score</code>         | Partial credit when a prediction lands in a documented synthetic source lineage or hybrid-related neighborhood.     |

The tests established the intended behavior rather than model superiority. They covered cases where the wrong species is still in the right genus, the wrong genus is still in the right family, synonym normalization collapses labels to the same accepted taxon, missing-rank bridges reduce inappropriate penalties, incoherent multi-rank predictions are counted, and reticulate source-lineage near misses receive bounded partial credit. This gave the later experiments a way to detect errors hidden by flat accuracy.

## Public WFO/GBIF/Open Tree Source-Plumbing Sample

The public-data-backed sample tested whether the schema could preserve real source-specific identifiers and statuses without forcing premature reconciliation. The builder `scripts/build_public_taxonomy_sample.py` writes `data/public_taxonomy_sample/v0.1/` with `seed_names.csv`, raw WFO/GBIF/Open Tree JSON responses, normalized `taxa.csv`, `names.csv`, `source_crosswalk.csv`, `hyperedges.csv`, `splits.csv`, `metadata.json`, and `source_coverage.png`.

The sample contained 16 seed names. All 16 had at least two source matches, and the validation record reports that 15 had matches from all three sources. The frozen raw cache contained 48 JSON responses. The metadata recorded access date 2026-05-17, endpoint URLs, query parameters, citation notes, limitations, software versions, and hashes.

The three source endpoints were:

| Source    | Endpoint                                                                                                              |
|-----------|-----------------------------------------------------------------------------------------------------------------------|
| WFO       | <a href="https://list.worldfloraonline.org/matching_rest.php">https://list.worldfloraonline.org/matching_rest.php</a> |
| GBIF      | <a href="https://api.gbif.org/v1/species/match">https://api.gbif.org/v1/species/match</a>                             |
| Open Tree | <a href="https://api.opentreeoflife.org/v3/tnrs/match_names">https://api.opentreeoflife.org/v3/tnrs/match_names</a>   |

The normalized tables contained 16 seed-name rows, 62 source-specific taxon rows, 63 name

rows, 16 source-crosswalk rows, 327 hyperedge rows, and 47 split rows. Public hyperedges were limited to the evidence families supported by the sample: 60 taxonomic\_parentage rows, 110 regional\_checklist\_context rows, and 157 synonym\_cluster rows. The public sample emitted no reticulate\_or\_hybrid\_signal hyperedges.

The main result was source preservation. `source_crosswalk.csv` kept WFO identifiers, GBIF keys, Open Tree OTT identifiers, source statuses, and source-specific match behavior visible instead of collapsing them into one accepted record. All 16 seeds had disagreement category `different_status`, largely because WFO matching returned candidate-level records while GBIF returned `ACCEPTED` matches and Open Tree returned `TNRS` matches. The report treats this as a representational source-status difference, not as evidence of biological disagreement.

Leakage controls were included so the public sample could support later experiments without becoming a hidden answer key. Accepted-name mappings were marked `task_visibility=name_normalization_only`, and split groups did not cross splits. This made the sample useful for real public source plumbing while preserving the boundary between name normalization and broader classification tasks.

The public sample’s limits are part of the result. It supports real name, identifier, rank, status, taxonomy, synonym, and source-context transformations. It does not support real trait syndromes, occurrence-quality conclusions, geographic ranges, reticulation or hybrid-origin claims, or phylogenetic novelty. Those remained synthetic benchmark mechanisms or future external-data questions.

## Baseline, Hypergraph, And Ablation Results

The direct model comparison used the synthetic benchmark and the public WFO/GBIF/Open Tree sample in different roles. The synthetic benchmark supplied the scored examples and known mechanisms. The public sample was used as a source-plumbing check, not as evidence for real trait, reticulation, occurrence-quality, range, or phylogenetic novelty claims.

The baseline suite compared label-frequency, taxonomy-depth, flat lookup or centroid scoring, tree/DAG scoring, ordinary graph scoring, clique expansion, and native role-labeled hypergraph scoring. The runner was `scripts/run_synthetic_experiments.py`, the scoring library was `tools/baselines.py`, and the experiment outputs were written under `data/experiments/synthetic_v0.1/`. The design constrained each baseline by information budget: the tree/DAG baseline used accepted taxonomic parentage, the ordinary graph baseline used pairwise typed relations, clique expansion converted hyperedges to pairwise links, and native hypergraph scoring kept role-labeled multi-way incidence.

The main test split contained 22 examples. On flat exact match, the structure-aware methods tied:

| Model family      | Flat exact match | Synonym-normalized exact match | Mean hierarchy distance |
|-------------------|------------------|--------------------------------|-------------------------|
| Tree/DAG          | 0.409091         | 0.409091                       | 2.181818                |
| Ordinary graph    | 0.409091         | 0.409091                       | 2.181818                |
| Clique expansion  | 0.409091         | 0.409091                       | 2.090909                |
| Native hypergraph | 0.409091         | 0.409091                       | 1.772727                |

This result is the central empirical answer. Native hypergraph scoring improved hierarchy-

aware distance, but it did not improve exact accuracy. The measurable advantage was therefore about how far wrong the prediction was in the hierarchy, not about producing more exact accepted-taxon matches on this split.

The case-type breakdown narrowed the interpretation. Strict hierarchy cases were solved by all structure-aware methods. Reticulate exact accuracy tied at 0.750000; native hypergraph improved reticulate hierarchy distance to 0.750000, compared with 1.000000 for tree/DAG, ordinary graph, and clique expansion. Missing-rank evaluation had only one held-out case, which all structure-aware methods solved. The held-out split had no noisy-occurrence or trait-convergence examples, so the experiment could not validate predictive effects for those edge families.

The ablations shaped the claim boundary:

| Ablation or perturbation                    | Native hypergraph effect                                                                           |
|---------------------------------------------|----------------------------------------------------------------------------------------------------|
| Collapse native scoring to clique expansion | Hierarchy distance worsened by +0.318182, moving native to 2.090909.                               |
| Remove missing-rank bridges                 | Hierarchy distance worsened by +0.227273.                                                          |
| Randomize accepted-name mappings            | Hierarchy distance worsened by +1.227273; exact and synonym-normalized accuracy fell by -0.363636. |
| Remove or randomize trait edges             | No aggregate effect in this test split.                                                            |
| Remove reticulate edges                     | No aggregate effect for native hypergraph in aggregate.                                            |
| Remove synonym hyperedges                   | No aggregate effect, while accepted-name mapping randomization had a large effect.                 |

The accepted-name mapping sensitivity is important. It shows that name and label plumbing explained substantial signal. The null synonym-edge ablation prevents a broad claim that synonym hyperedges alone drove the result. The null trait and reticulate aggregate ablations also prevent strong claims that those families were robust predictive drivers in this split.

The strict negative control checked whether the benchmark was unfair to tree/DAG methods. When synonym, missing-rank, reticulation, trait-convergence, and occurrence-noise mechanisms were disabled, tree/DAG and native hypergraph both had mean hierarchy distance 0.571429. This satisfied the intended control: when the data were tree-generated, the tree/DAG baseline was not structurally disadvantaged.

The corrected experiment therefore supports a narrow, hierarchy-aware native-hypergraph benefit on a deterministic synthetic benchmark. It does not support broad native-hypergraph predictive superiority, and it does not support new biological conclusions about real plant taxa.

## Formal Clique-Expansion Diagnostic

The formal diagnostic addressed a representational question raised by the experiment: when does replacing a hyperedge with pairwise graph edges preserve the meaning of the original evidence? The answer was expressed as a finite proposition and verifier in docs/formal\_diagnostic.md, scripts/verify\_formal\_diagnostic.py, tests/test\_formal\_diagnostic.py, and outputs under data/formal\_diagnostic/.

For a hyperedge with  $k$  taxon members, clique expansion creates:



$$\frac{k(k-1)}{2}$$

unordered taxon pairs. The formula itself is elementary. The diagnostic contribution is the semantic safety condition: clique expansion is safe only when every introduced pairwise adjacency is licensed by the hyperedge family for the task.

The condition matters because co-membership is not always pairwise similarity. A regional checklist context can connect several taxa, a source, and a region without claiming all taxa are mutually similar. A trait-syndrome hyperedge can be a convergence stressor rather than a taxonomic-closeness claim. A reticulate or hybrid-like hyperedge can contain source-lineage roles where some pairwise relations are meaningful and others are not.

The finite verifier checked the pair-count formula for representative arities:

| Taxon members $k$ | Introduced unordered pairs |
|-------------------|----------------------------|
| 0                 | 0                          |
| 1                 | 0                          |
| 2                 | 1                          |
| 3                 | 3                          |
| 5                 | 10                         |

The diagnostic was then linked back to the experiment through the M6 clique diagnostic summary:

| Hyperedge family                    | Hyperedges | Introduced pairs | Licensed pairs | Unlicensed ratio |
|-------------------------------------|------------|------------------|----------------|------------------|
| regional_<br>checklist_<br>context  | 4          | 40               | 0              | 1.000000         |
| reticulate_<br>or_hybrid_<br>signal | 7          | 21               | 14             | 0.333333         |
| trait_<br>syndrome                  | 22         | 22               | 0              | 1.000000         |

These numbers explain why clique expansion can be misleading even when it is computationally convenient. In the reported synthetic benchmark, regional checklist context and trait-syndrome hyperedges created pairwise taxon adjacencies that were not licensed as taxonomic similarity. Reticulate signal hyperedges were mixed: some source-lineage pairs were licensed, while others were not.

The diagnostic does not prove that clique expansion is always worse, and it does not prove that native hypergraphs are generally more accurate. It proves a conditional warning: pairwise expansion preserves semantics only when the task’s evidence policy licenses every introduced adjacency. That warning directly supports the report’s narrower conclusion that the native hypergraph representation is useful for preserving role-labeled, multi-way distinctions even when predictive accuracy gains are limited.

## Claim Classification And Hypothesis Assessment

The research output separates modeling claims from biological claims. This separation is necessary because the campaign combined real public source plumbing with synthetic stress tests. The real WFO/GBIF/Open Tree sample supports name, identifier, status, rank, synonym, taxonomy, and source-context transformations. The reticulation, trait-convergence, noisy-occurrence, and missing-rank stressors used for scored model comparison are synthetic benchmark mechanisms.

The validated modeling claims are concrete. First, the formal schema shows that plant-taxonomy evidence can be represented as a typed, role-labeled hypergraph without collapsing taxonomy, phylogeny, nomenclature, trait similarity, occurrence evidence, or dataset labels into one evidence layer. The schema's value is not just that it permits multi-way edges; it defines what each hyperedge family is allowed and forbidden to prove.

Second, the synthetic benchmark provides a deterministic test bed for strict hierarchy, synonym or renamed labels, missing ranks, trait convergence, noisy occurrence, and reticulate or hybrid-like stressors. Because the correct mechanisms are known, the benchmark can be used to ask falsifiable questions about when a tree/DAG, ordinary graph, clique expansion, or native hypergraph should succeed or fail.

Third, the metric stack exposes information that flat exact accuracy hides. In the M6 experiment, tree/DAG, ordinary graph, clique expansion, and native hypergraph scoring all tied at 0.409091 flat exact match and 0.409091 synonym-normalized exact match. The model ranking changed only under hierarchy distance, where native hypergraph scoring reached 1.772727 compared with 2.090909 for clique expansion and 2.181818 for tree/DAG and ordinary graph. The measurable advantage was therefore hierarchy-aware, not an exact-label accuracy gain.

Fourth, the public WFO/GBIF/Open Tree sample validates narrow source plumbing. The sample froze source responses for 16 seed names, preserved source-specific identifiers and statuses, and transformed the records into normalized tables and hyperedges. Its different\_status category reflects source and matching-mode differences, not biological conflict.

Fifth, the clique-expansion diagnostic validates a conditional semantic warning. A  $k$ -taxon hyperedge expands to  $k(k - 1) / 2$  unordered taxon pairs under clique expansion, and that conversion is safe only when every introduced pairwise adjacency is licensed by the hyperedge family's declared meaning. The M6 diagnostic showed unlicensed introduced-pair ratios of 1.000000 for regional\_checklist\_context, 0.333333 for reticulate\_or\_hybrid\_signal, and 1.000000 for trait\_syndrome.

Several stronger claims were weakened or not supported. Broad native-hypergraph predictive superiority was not established, because exact accuracy did not improve over tree/DAG, ordinary graph, or clique expansion. The reticulation advantage was weakened: native hypergraph scoring improved reticulate hierarchy distance, but reticulate exact accuracy tied at 0.750000, and removing reticulate hyperedges had zero aggregate effect for native hypergraph in the M6 split. Trait and occurrence predictive claims were not validated, because the held-out test split lacked noisy-occurrence and trait-convergence examples and trait-edge ablations had no aggregate effect. A synonym-edge-specific performance claim was also not supported: removing synonym hyperedges had no aggregate effect, while randomizing accepted-name mappings had a large effect.

The public-data-limited claims are narrower than the synthetic claims. WFO, GBIF, and Open Tree can support a small reproducible source-plumbing prototype for names, identifiers, statuses, taxonomy, synonymy, and source contexts. They did not, in this campaign, support real trait analysis, occurrence-quality analysis, hybrid-origin evidence, reticulation evidence, or phylogenetic novelty.

Unsupported biological claims were explicitly avoided. The report does not claim new plant taxonomy, corrected accepted names, new species relationships, real hybrid origins, real trait syndromes, species ranges, checklist truth, occurrence-quality conclusions, or phylogenetic novelty. WFO, GBIF, and Open Tree are treated as distinct evidence systems rather than reconciled into one biological authority.

The final hypothesis assessment follows from those claim boundaries:

| Hypothesis                                                                                                                                           | Assessment                                      | Basis                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| H1: synonym-normalized hierarchy metrics will change model rankings compared with flat accuracy.                                                     | Weakly supported.                               | The hierarchy-aware metric changed the interpretation, but synonym-normalized exact-match rankings matched flat exact-match rankings because predictions were accepted taxon IDs. |
| H2: hypergraph models will outperform tree/DAG baselines on synthetic reticulate and missing-rank cases, but not necessarily strict hierarchy cases. | Weakened.                                       | Native hypergraph scoring improved overall and reticulate hierarchy distance, but exact accuracy tied and the remove-reticulate ablation was flat in aggregate.                   |
| H3: apparent hypergraph gains may vanish when synonym normalization and label-frequency baselines are included.                                      | Supported in part.                              | Randomizing accepted-name mappings sharply degraded performance, showing that name and label plumbing explained substantial signal.                                               |
| H4: clique expansion will sometimes create false pairwise similarity from multi-way evidence.                                                        | Supported as a conditional semantic diagnostic. | The formal diagnostic and M6 linkage identify when introduced pairwise adjacencies are unlicensed by the original hyperedge semantics.                                            |
| H5: the most durable contribution may be a benchmark and metric rather than a predictive model.                                                      | Strongly supported.                             | The validated durable outputs are the benchmark, metric stack, public-data plumbing, ablation suite, and formal diagnostic.                                                       |

The final audit headline was 8 validated · promise\_check=green: all eight planned milestones were validated with high confidence, and no critical, moderate, or minor final-audit findings were recorded. That validation status means the deliverables are internally complete for the scoped campaign. It does not convert synthetic mechanisms into biological evidence or turn a hierarchy-distance gain into broad predictive superiority.

## Limitations, Residual Debt, And Future Work

The final audit recorded no blocking residual debt and no outstanding final-audit findings. This means the campaign’s promised artifacts and checks were considered complete within

scope. It does not mean that the scientific question is exhausted, or that later work should treat the synthetic results as a settled biological conclusion.

The main limitations are evidence limitations. Public reticulation evidence was absent, so all reticulate and hybrid-like cases used in scored experiments are synthetic. Public trait evidence was absent; TRY access and redistribution constraints made it unsuitable for a no-auth redistributable first artifact, and BIEN/RBIEN was not locally smoke-tested. Public occurrence provenance was absent from the transformed M5 sample, so occurrence-quality and geography claims remain untested.

The experimental limitations are also explicit. M6 used deterministic scoring baselines, not optimized classifiers, learned hypergraph neural models, or large-scale trained systems. The synthetic test split contained only 22 examples. It had no held-out noisy-occurrence or trait-convergence examples, and the missing-rank evaluation had only one held-out missing-rank case. The strict negative control showed that tree/DAG and native hypergraph both reached mean hierarchy distance 0.571429 when tree-generated data removed the stressors that could favor hypergraphs.

The public sample should not be overread. It is small, curated, and designed for source-specific plumbing rather than representative global plant-taxonomy inference. Its source-status differences are differences in provider status fields and matching behavior, not direct evidence of biological disagreement. WFO identifiers, GBIF taxon keys, and Open Tree OTT IDs remain source-specific unless a task-specific reconciliation rule is defined.

The artifact limitations are mostly reproducibility details. Stable hash validation focuses on CSV and JSON outputs, because PNG byte hashes may vary across matplotlib environments. Existing nonblocking process warnings in the root deliverables were not scientific blockers and did not change the reported results.

Recommended next work follows the audit report rather than the final-audit residual-debt list, which was empty:

1. Generate a larger stratified synthetic split that guarantees held-out trait-convergence, noisy-occurrence, missing-rank, and reticulate cases.
2. Add a stronger pairwise edge-typed graph baseline that preserves edge types and role labels as far as a graph representation can.
3. Add a public GBIF occurrence sample only if licensing, citation, caching, and reproducibility constraints are handled.
4. Test external transfer if a suitable plant taxonomy, name-normalization, or biodiversity label dataset is found.
5. Vary reticulation, missing-rank, trait-convergence, class imbalance, and occurrence-noise rates to map where native hypergraph structure helps or fails.

The practical next experiment is therefore not to scale the current claim, but to stress it. The benchmark and diagnostic framework make that possible: future work can ask where native hypergraphs retain an advantage, where fair graph baselines close the gap, and where the observed signal is mostly name normalization or taxonomy-depth structure.

## **Reproducibility And Artifact Map**

The campaign is reproducible through a small set of scripts, frozen data directories, tests, and report artifacts. The artifacts are organized around eight validated milestones, all with high-confidence final audit status.

| Milestone | Main reproducibility artifacts                                                                                                                                                         | What they reproduce or inspect                                                                               |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| M1        | docs/data_feasibility_map.md, data/source_feasibility_matrix.csv, data/source_probe_results.json, scripts/probe_public_sources.py                                                      | Source feasibility, access/citation notes, and small WFO/GBIF/Open Tree probes.                              |
| M2        | docs/hypergraph_schema.md                                                                                                                                                              | Typed hypergraph schema, hyperedge permissions, leakage controls, and baseline information budgets.          |
| M3        | scripts/generate_synthetic_benchmark.py, docs/synthetic_benchmark_design.md, data/synthetic_benchmark/v0.1/                                                                            | Deterministic synthetic taxonomy benchmark, tables, metadata, and composition figure.                        |
| M4        | tools/hierarchy_metrics.py, tests/test_hierarchy_metrics.py, tests/test_synthetic_benchmark.py                                                                                         | Hierarchy-aware metrics and benchmark integrity checks.                                                      |
| M5        | scripts/build_public_taxonomy_sample.py, tools/source_sample_checks.py, tests/test_public_taxonomy_sample.py, docs/public_taxonomy_sample_design.md, data/public_taxonomy_sample/v0.1/ | Frozen public WFO/GBIF/Open Tree name, identifier, taxonomy, source-crosswalk, and hyperedge sample.         |
| M6        | tools/baselines.py, scripts/run_synthetic_experiments.py, tests/test_baselines.py, docs/m6_experiment_report.md, data/experiments/synthetic_v0.1/                                      | Baseline comparisons, ablations, strict negative control, public-plumbing check, result tables, and figures. |
| M7        | scripts/verify_formal_diagnostic.py, tests/test_formal_diagnostic.py, docs/formal_diagnostic.md, data/formal_diagnostic/                                                               | Clique-expansion pair-count verification, semantic-safety diagnostic, and M6 diagnostic summary.             |
| M8        | final_report.md, artifact_index.md, research_contribution_ledger.md, audit_report.md                                                                                                   | Final synthesis, artifact map, contribution ledger, audit judgment, and next-experiment list.                |

The synthetic benchmark can be regenerated with:

```
python3 scripts/generate_synthetic_benchmark.py --seed 20260517 --out-dir data/synthetic_benchmark/v0.1/
```

The public WFO/GBIF/Open Tree source sample can be rebuilt from the frozen cache, or from

network requests when access conditions allow, with:

```
python3 scripts/build_public_taxonomy_sample.py --out-dir data/public_taxonomy_sample/v0.1
```

The baseline, hypergraph, ablation, strict-control, and figure outputs can be regenerated with:

```
python3 scripts/run_synthetic_experiments.py --benchmark-dir data/synthetic_benchmark/v0.1 --public-sample-dir
↳ data/public_taxonomy_sample/v0.1 --out-dir data/experiments/synthetic_v0.1 --seed 20260517 --ablation all
```

The formal diagnostic can be regenerated from the M6 clique false-similarity table with:

```
python3 scripts/verify_formal_diagnostic.py --m6-diagnostic
↳ data/experiments/synthetic_v0.1/clique_false_similarity.csv --out-dir data/formal_diagnostic
```

The primary validation commands are:

```
python3 -m py_compile scripts/generate_synthetic_benchmark.py scripts/build_public_taxonomy_sample.py
↳ scripts/run_synthetic_experiments.py scripts/verify_formal_diagnostic.py
python3 -m unittest tests/test_synthetic_benchmark.py tests/test_hierarchy_metrics.py
↳ tests/test_public_taxonomy_sample.py tests/test_baselines.py tests/test_formal_diagnostic.py
python3 -m long_exposure.tools.promise_check .
```

The main generated data and figure outputs are:

| Output group          | Files                                                                                                                                                                                                                                                                                  |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Synthetic benchmark   | data/synthetic_benchmark/v0.1/taxa.csv, names.csv, hyperedges.csv, examples.csv, splits.csv, metadata.json, composition.png                                                                                                                                                            |
| Public source sample  | data/public_taxonomy_sample/v0.1/seed_names.csv, taxa.csv, names.csv, source_crosswalk.csv, hyperedges.csv, splits.csv, metadata.json, source_coverage.png, and 48 raw JSON responses under raw/                                                                                       |
| Synthetic experiments | data/experiments/synthetic_v0.1/results.csv, predictions.csv, ablation_results.csv, case_type_breakdown.csv, clique_false_similarity.csv, strict_negative_control.csv, summary.json, metric_comparison.png, ablation_heatmap.png, case_type_breakdown.png, clique_false_similarity.png |
| Formal diagnostic     | data/formal_diagnostic/finite_examples.csv, m6_clique_diagnostic_summary.csv, formal_diagnostic_summary.json, clique_warning_diagnostic.png                                                                                                                                            |

The reproducibility boundary is deliberate. CSV and JSON artifacts carry the stable tabular and metadata evidence for the reported findings. PNG figures are included for interpretation, but byte-level PNG identity may vary across plotting environments.

## Conclusions

Plant-taxonomy evidence can be represented as a role-labeled hypergraph in a way that preserves distinctions that trees, directed acyclic graphs, ordinary pairwise graphs, and flat labels tend to collapse. The cleanest formulation produced here keeps taxonomy, phylogeny, nomenclature, trait similarity, occurrence evidence, and dataset labels as distinct evidence layers, then connects them through typed hyperedge families with explicit allowed and forbidden inferences.

The measured advantage was real but narrow. On the deterministic synthetic benchmark, native hypergraph scoring improved mean hierarchy distance to 1.772727, compared with 2.090909 for clique expansion and 2.181818 for tree/DAG and ordinary graph. It did not improve flat exact match or synonym-normalized exact match, where the structure-aware methods tied at 0.409091. The result supports hierarchy-aware and diagnostic value, not broad predictive superiority.

The falsification results are part of the contribution. Accepted-name mapping randomization had a large effect, showing that name and label plumbing explained substantial signal. Removing synonym hyperedges, trait edges, and reticulate edges did not produce broad aggregate effects in the M6 split. The strict negative control showed that tree/DAG and native hypergraph scoring tied when the data were tree-generated. Those results constrain the claim and make the benchmark useful for later work.

The public-data-backed component supports source plumbing rather than biological discovery. WFO, GBIF, and Open Tree records were transformed into a reproducible source-specific name, identifier, taxonomy, and status sample, but the campaign did not validate real plant reticulation, trait syndromes, occurrence-quality patterns, species ranges, checklist truth, or phylogenetic novelty.

The formal diagnostic gives the clearest general lesson. A hyperedge with  $k$  taxon members becomes  $k(k - 1) / 2$  pairwise taxon adjacencies under clique expansion, and the conversion is semantically safe only when all of those adjacencies are licensed for the task. In the M6 diagnostic, several hyperedge families created unlicensed pairwise claims under clique expansion, explaining why native hypergraph representation can matter even when flat accuracy does not improve.

The final answer is therefore scoped: hypergraph modeling is worth pursuing for plant classification and taxonomic conflict analysis when the evidence is multi-way, source-scoped, role-dependent, or evaluated with hierarchy-aware metrics. The durable result of this campaign is a reusable benchmark, metric stack, public-source transformation, ablation suite, and formal diagnostic that make that claim testable.

## References

- [1] World Flora Online, “WFO Plant List API,” World Flora Online, 2026. <https://list.worldfloraonline.org/index.php> (accessed 2026-05-17).
- [2] World Flora Online, “Name Matching REST API,” World Flora Online, 2026. [https://list.worldfloraonline.org/matching\\_rest.php](https://list.worldfloraonline.org/matching_rest.php) (accessed 2026-05-17).
- [3] World Flora Online, “GraphQL API,” World Flora Online, 2026. [https://list.worldfloraonline.org/graphql\\_index.php](https://list.worldfloraonline.org/graphql_index.php) (accessed 2026-05-17).
- [4] The World Flora Online Consortium et al., “World Flora Online Plant List December 2025,” Zenodo, 2025. <https://doi.org/10.5281/zenodo.18007552> (accessed 2026-05-17).

- [5] World Flora Online, "The WFO Plant List," World Flora Online, 2026. <https://about.worldfloraonline.org/plant-list/> (accessed 2026-05-17).
- [6] GBIF Secretariat, "GBIF API Reference," GBIF Technical Documentation, 2026. <https://techdocs.gbif.org/en/openapi/> (accessed 2026-05-17).
- [7] GBIF Secretariat, "Species API," GBIF Technical Documentation, 2026. <https://techdocs.gbif.org/en/openapi/v1/species> (accessed 2026-05-17).
- [8] GBIF Secretariat, "Occurrence API," GBIF Technical Documentation, 2026. <https://techdocs.gbif.org/en/openapi/v1/occurrence> (accessed 2026-05-17).
- [9] GBIF Secretariat, "API Downloads," GBIF Technical Documentation, 2026. <https://techdocs.gbif.org/en/data-use/api-downloads> (accessed 2026-05-17).
- [10] GBIF Secretariat, "Citation Guidelines," GBIF, 2026. <https://www.gbif.org/citation-guidelines> (accessed 2026-05-17).
- [11] GBIF Secretariat, "Terms of Use," GBIF, 2026. <https://www.gbif.org/terms> (accessed 2026-05-17).
- [12] Open Tree of Life, "Link to current Open Tree API," Open Tree of Life, 2026. <https://opentreeoflife.github.io/develop/api> (accessed 2026-05-17).
- [13] OpenTreeOfLife, "TNRS API v3," Open Tree of Life germinator wiki, 2022. <https://github.com/OpenTreeOfLife/germinator/wiki/TNRS-API-v3> (accessed 2026-05-17).
- [14] Open Tree of Life, "Licenses," Open Tree of Life, 2026. <https://tree.opentreeoflife.org/about/licenses> (accessed 2026-05-17).
- [15] Cody E. Hinchliff, Stephen A. Smith, James F. Allman, et al., "Synthesis of Phylogeny and Taxonomy into a Comprehensive Tree of Life," Proceedings of the National Academy of Sciences, 2015. <https://doi.org/10.1073/pnas.1423041112> (accessed 2026-05-17).
- [16] BIEN, "RBIEN," Botanical Information and Ecology Network, 2026. <https://bien.nceas.ucsb.edu/bien/tools/rbien/> (accessed 2026-05-17).
- [17] Brian Maitner, "BIEN: Tools for Accessing the Botanical Information and Ecology Network Database," R-universe package manual, version 1.2.7, 2026. <https://cran.r-universe.dev/BIEN/doc/manual.html> (accessed 2026-05-17).
- [18] TRY, "TRY Plant Trait Database," TRY, 2026. <https://www.try-db.org/TryWeb/Home.php> (accessed 2026-05-17).
- [19] TRY, "Intellectual Property Guidelines for the TRY Plant Trait Database," TRY, updated 2023. [https://www.try-db.org/TryWeb/TRY\\_Intellectual\\_Property\\_Guidelines.pdf](https://www.try-db.org/TryWeb/TRY_Intellectual_Property_Guidelines.pdf) (accessed 2026-05-17).
- [20] TRY, "Free Access TRY Data," TRY, 2026. [https://www.try-db.org/TryWeb/TRY\\_Policy\\_Free\\_Access\\_Data.pdf](https://www.try-db.org/TryWeb/TRY_Policy_Free_Access_Data.pdf) (accessed 2026-05-17).
- [21] Dengyong Zhou, Jiayuan Huang, and Bernhard Scholkopf, "Learning with Hypergraphs: Clustering, Classification, and Embedding," Advances in Neural Information Processing Systems 19, 2006. <https://papers.nips.cc/paper/3128-learning-with-hypergraphs-clustering-classification-and-embedding> (accessed 2026-05-17).
- [22] Sameer Agarwal, Kristin Branson, and Serge Belongie, "Higher Order Learning with Graphs," International Conference on Machine Learning, 2006. <https://doi.org/10.1145/1143844.1143847> (accessed 2026-05-17).



- [23] Carlos N. Silla Jr. and Alex A. Freitas, "A Survey of Hierarchical Classification across Different Application Domains," *Data Mining and Knowledge Discovery*, 2011. <https://doi.org/10.1007/s10618-010-0175-9> (accessed 2026-05-17).
- [24] Roderic D. M. Page, "Ozymandias: A Biodiversity Knowledge Graph," *PeerJ*, 2019. <https://peerj.com/articles/6739/> (accessed 2026-05-17).
- [25] Lyubomir Penev, Teodor Georgiev, Pavel Stoev, et al., "OpenBiodiv: A Knowledge Graph for Literature-Extracted Linked Open Data in Biodiversity Science," *Publications*, 2019. <https://www.mdpi.com/2304-6775/7/2/38> (accessed 2026-05-17).
- [26] Jens Kattge, Gerhard Boenisch, Sandra Diaz, et al., "TRY Plant Trait Database - Enhanced Coverage and Open Access," *Global Change Biology*, 2020. <https://doi.org/10.1111/gcb.14904> (accessed 2026-05-17).